

Mohamed Mahmoud Samy

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SUMMARY

Mechatronics engineering student at Ain Shams University with a proven track record across the full engineering stack, from hardwired relay logic and PCB design to autonomous navigation algorithms and FPGA development. Experienced through competitive robotics (FITENTH), aerospace internships (EgSA), and hands-on industrial projects spanning pneumatic systems, embedded control, and electromechanical integration.

EDUCATION

B.Sc. Mechatronics Engineering

Ain Shams University • 2027 • 3.22

EXPERIENCE

Autonomous Systems Engineer

FITENTH Racing team

September 2025 - Present, Ain Shams University

- Developed and tuned autonomous navigation and control algorithms for the race vehicle using LiDAR-based perception data.
- Implemented Follow-The-Gap (FTG), Pure Pursuit, and Model Predictive Control (MPC) algorithms for obstacle avoidance and trajectory tracking.
- Collaborated with localization and perception sub teams to integrate real-time autonomous driving architecture in ROS2.

Intern

EgSA-Egyptian Space Agency

January 2025 - February 2025, New Administrative Capital, Egypt • On-site

- Completed an intensive internship focused on FPGA design and VHDL development for embedded aerospace systems.
- Designed, tested, and implemented finite state machines (FSMs), hierarchical modules, and basic arithmetic logic components, Conducting RTL simulations, functional verification, and timing analysis.

Engineer Intern

Ghabbour

July 2024 - August 2024, Egypt • On-site

- Gained hands-on exposure to automotive mechanical systems, maintenance procedures, and fault diagnosis in a real service environment.
- Assisted in identifying and troubleshooting mechanical and electromechanical issues under the supervision of senior engineers, and participated in standardized repair, inspection, and preventive maintenance workflows.

CERTIFICATIONS

Python for Everybody

University of Michigan • 2026

FPGA & VHDL Workshop

EgSA-Egyptian Space Agency • 2025

SOLIDWORKS 3D CAD for Education

Dassault Systèmes • 2025

MATLAB & Simulink: Onramp + Modeling & Simulation

MathWorks • 2025

OSHA Construction & Industry Course

Orascom Construction PLC • 2025

SKILLS

Embedded Control: Embedded Systems, STM32, PLC, Control Systems, Electro-Pneumatics.

Electronics & Hardware: PCB Design, Circuit Simulation, Sensors & Actuators, Industrial Control Components.

Mechanical Design: SolidWorks, CAD, Mechanical Drawings, Power Transmission, Stress Analysis.

Programming & HDL: C/C++, Python, VHDL, MATLAB, Simulink